MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, PIN CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

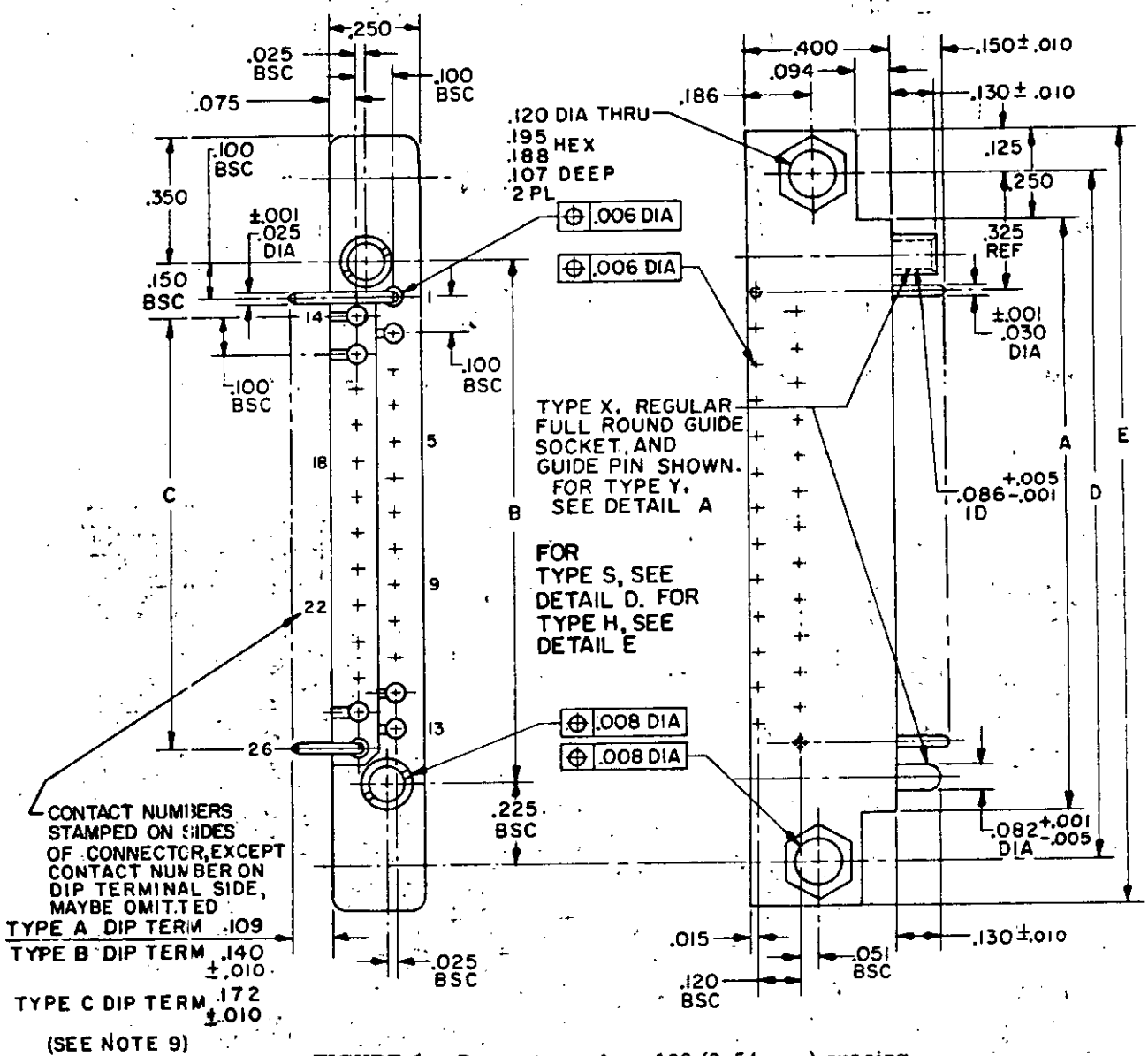
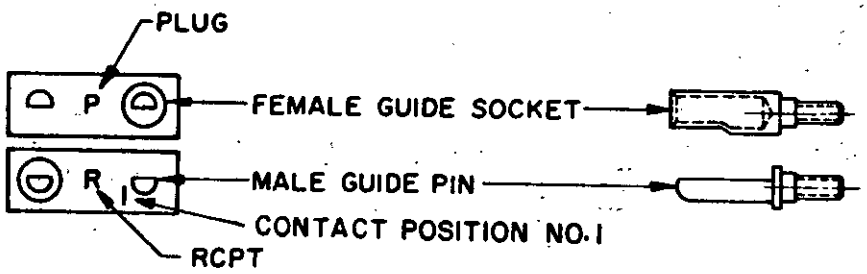


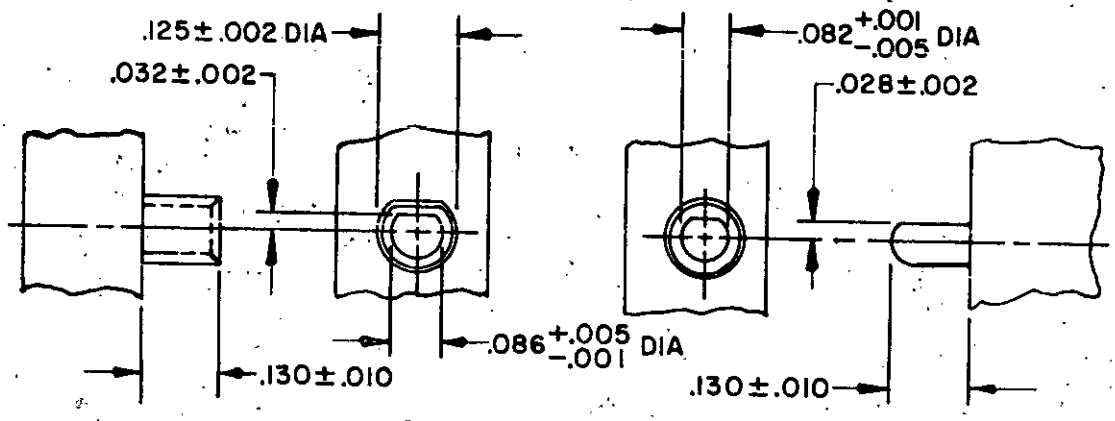
FIGURE 1. Connector, plug .100 (2.54 mm) spacing.

FSC 5935

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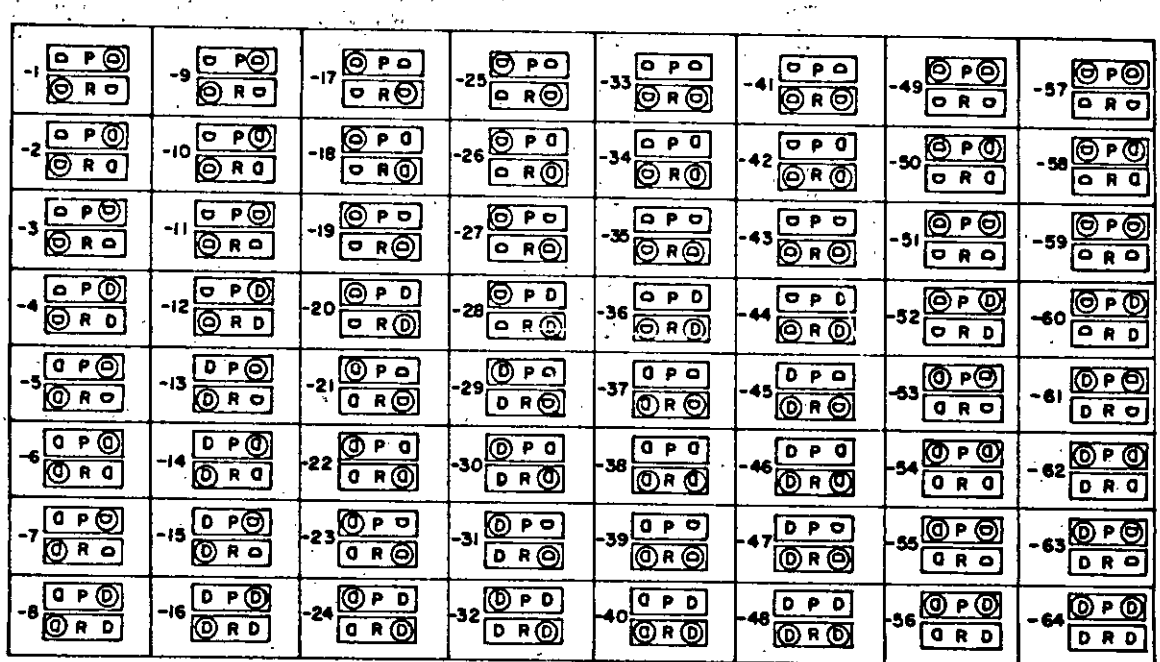


CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS. (TYPE Y)



Detail A. Polarization pins/socket. (For polarization configuration see detail B)

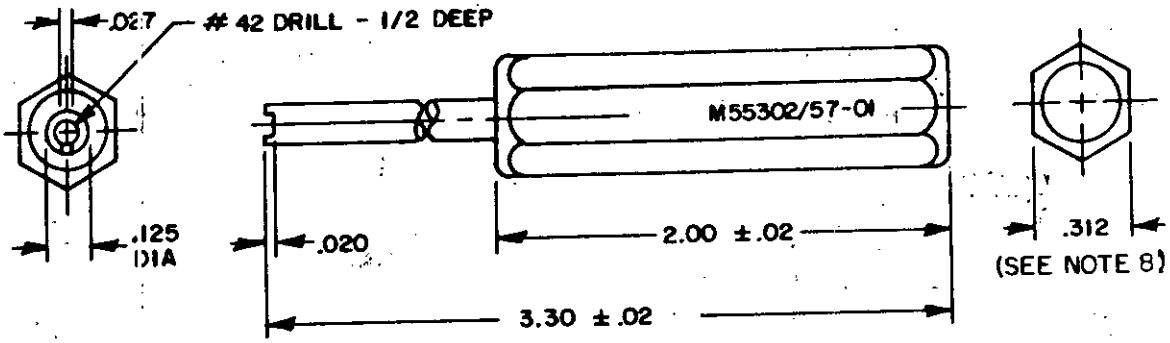
DETAIL B. Polarization configuration chart.



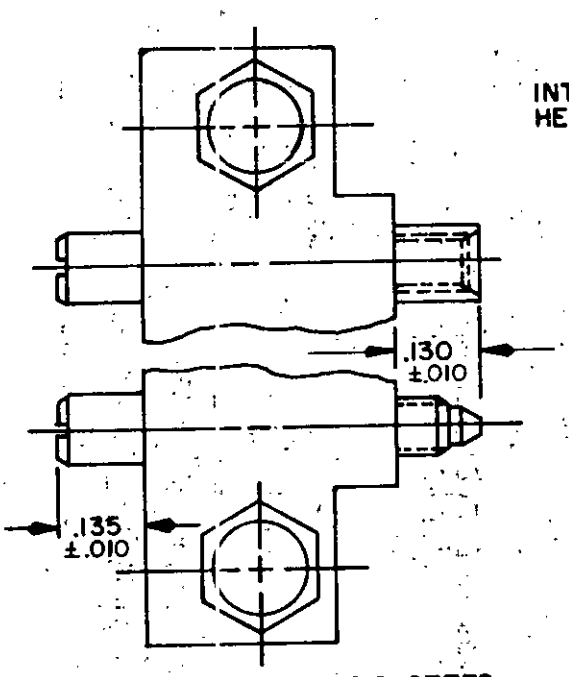
NOTE: Pins/sockets in chart can be easily adjusted to any desired configuration by the spanner wrench (see detail C).

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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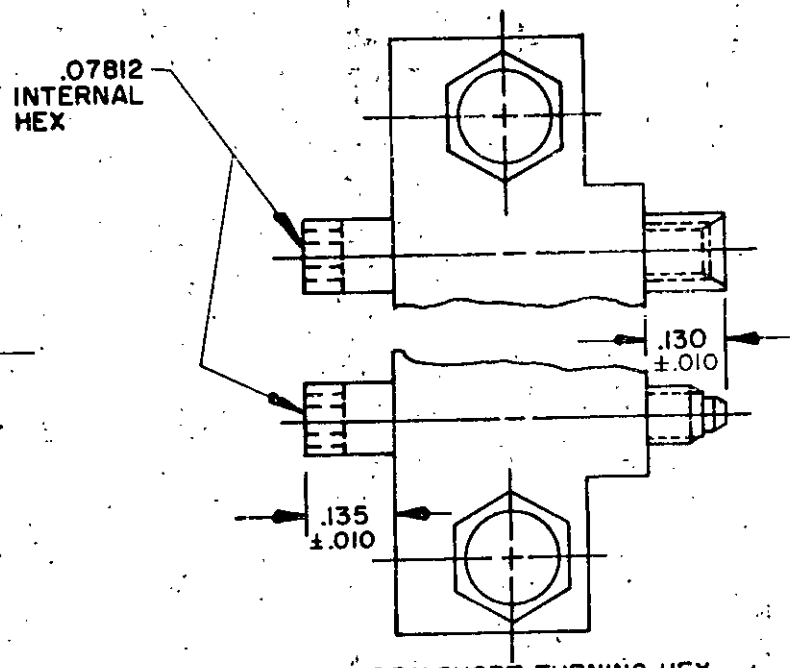


DETAIL C SPANNER WRENCH



TYPE S SHORT TURNING SLOTTED JACK SCREW

DETAIL D

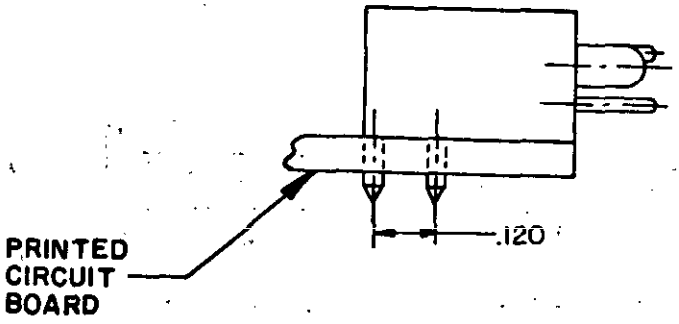


TYPE H SHORT TURNING HEX JACK SCREWS

DETAIL E

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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INCHES	MM	INCHES	MM
.001	.03	.030	.76
.002	.05	.032	.81
.005	.13	.051	1.30
.006	.15	.075	1.91
.008	.20	.07812	1.9842
.010	.25	.082	2.08
.015	.38	.086	2.18
.020	.51	.094	2.39
.025	.64	.100	2.54
.027	.69	.107	2.72
.028	.71		

Number of contacts	Dimensions				
	A ±.010	BSC B	REF C	BSC D	E ±.010
10	.850 (21.59)	.650 (16.51)	.400 (10.16)	1.100 (27.94)	1.350 (34.29)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)	1.300 (33.02)	1.550 (39.37)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)	1.600 (40.64)	1.850 (46.99)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)	1.800 (45.72)	2.050 (52.07)
26	1.650 (41.91)	1.450 (36.83)	1.200 (30.48)	1.900 (48.26)	2.150 (54.61)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)	2.100 (53.34)	2.350 (59.69)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)	2.400 (60.96)	2.650 (67.31)
40	2.350 (59.69)	2.150 (54.61)	1.900 (48.26)	2.600 (66.04)	2.850 (72.39)
44	2.550 (64.77)	2.350 (59.69)	2.100 (53.34)	2.800 (71.12)	3.050 (77.47)
50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)	3.100 (78.74)	3.350 (85.09)
54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)	3.300 (83.82)	3.550 (90.17)
56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)	3.400 (86.36)	3.650 (92.71)
60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)	3.600 (91.44)	3.850 (97.79)
66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)	3.900 (99.06)	4.150 (105.41)
70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)	4.100 (104.14)	4.350 (110.49)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/58, /62, /64, /65, and /66 using appropriate hardware.
5. Numbers indicating every fourth cavity are stamped on side except that the number one contact shall be marked.
6. Termination layout on .020 (0.51 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Handle shape optional.
9. Optional undercut .020 (0.51 mm) maximum, on PCB side for cleaning purposes.

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion resisting stainless steel per ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max. other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper per QQ-C-550, nickel plated per QQ-N-290, class 2, 50 to 150 microinches.

Plating: The contact plating shall be per MIL-C-55302 or gold per MIL-G-45204, type II, grade C, class 1, over nickel plating per QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold per MIL-G-45204, type II, grade C, class 00, over nickel plating per QQ-N-290, class 02, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Contact engagement and separation forces (No. 22 contacts):
Minimum separation: 0.5 ounce.
Maximum engagement: 12.0 ounces.

Mating and unmating:
Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum per contact.

Polarization: Type X, regular, guide pin and socket, full round (see figure 1). Type Y, adjustable "D" shaped guide pin and/or socket (see figure 1, details A and B). All connectors are supplied in the -1 polarized position. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three (3) spaces after the part number on the purchase order. Example M55302/57-A10Y [-32]. Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench (M55302/57-01). Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing an "-0" in brackets three (3) spaces after the part number on the purchase order.

Example of part number:

M55302/57	L -	A	10	X
Basic number of specification sheet	Type of contacts	Type of terminal (see figure 1)	Number of contracts (see figure 1)	Type of mounting hardware (see details A,B, C,D,E of figure 1)
	L = Low insertion force contacts S = Standard insertion force contacts	A = dip terminal .109 long B = dip terminal .140 long C = dip terminal .172 long		Type X = full round Type Y = "D" shaped Type S = .135 slotted turning jackscrew Type H = .135 hex turning jackscrew

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS
Air Force - 19

Agent :
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3411-3)

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